Shashank Shaga

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EDUCATION

B.S Computer Science (Herbert College of Engineering) May 2028
University of Florida **GPA: 4.0**

PROFESSIONAL EXPERIENCE

Gator AI – *Machine Learning Engineer* Jan 2026 – Present

- Building a high-fidelity **chaotic dynamics simulation** of a double pendulum using **C++** and numerical solver(5k45), generating **6M+ trajectory data points** to train and validate neural networks.
- Using **Hiper-Gator** (supercomputer) cluster to construct **chaos maps** and compute Lyapunov exponents for nonlinear system sensitivity analysis using **parquet**.
- Developing **neural networks** architecture to approximate chaotic state transition reducing prediction **error by 35%** compared to baseline numerical extrapolation and enabling scalable AI predicted chaos forecasting.

Machine Intelligence Laboratory – *Software Programmer* Aug 2025 – Jan 2026

- Advanced underwater robot navigation efficiency by targeting a **30% reduction in travel distance** through ROS-based path optimization.
- Developing and testing **underwater computer vision models** to improve RGB spectrum recognition in challenging aquatic environments with rigorous testing to ensure 100 % autonomous route accuracy.
- Led the development of Navigator website redesign using **React + JS**, integrating **Figma designs**.

Society of Asian Scientists and Engineers (SASE) – *Software Engineer* Aug 2025 – Present

- Built a fully responsive website using **React, Tailwind, and Vite**, optimizing desktop load time by **25%**
- Implemented **10 + interactive UI animations and an AI chatbot**, improving user engagement by 30%.
- Integrated secure financial **API**'s, handling transactions, while also ensuring ADA accessibility by giving the user ability to adjust 4 key design elements meeting official APA guidelines.

TECHNICAL SKILLS

C++ | Python | Java | JavaScript | React | Next.js | SQL | Git | Agile | Tailwind CSS | Inventor | Fusion 360 |

PROJECT EXPERIENCE

Intel – *AI Interview Advisor* Jan 2026 – Present

- Built an AI-powered interview coaching system using **OpenAI Whisper**, **Docker** and Python-based resume parsing, generating **personalized feedback** and storing user performance metrics in an **SQLite database.**
- Developed a cross-platform mobile app with **React Native (TypeScript, Expo)** enabling real-time speech capture, feedback delivery, and contact storage for future interview follow-ups.

AI Advisor – *Web development Team* *Sep 2025 – Dec 2025*

- Built an AI-powered academic advising system using **Python and OpenAI's API**, analyzing students' degree requirements and history with **98% schedule-accuracy** to keep them on track for graduation.
- Designed a responsive dashboard using **React + Tailwind** that generates semester plans in **5 seconds**

Dance Sight APP – *Passion Project* Jun 2025 - Present

- Developing a React Native app that uses AI vision to track dancers' movements and provide real-time technique feedback, improving practice efficiency by 30% saving dancers 10+ hours each week.

ADDITIONAL INFORMATION

Certifications: IT specialists **(Java, JavaScript**), PMI **Project Management** Ready, RECF programming
Extracurriculars: 9+ years competitive dancer, Freelance Web developer, 1600+ rated Chess.